

PART 1 — Original Dataset EDA

First-pass exploration of the 6 core Epoch AI datasets: overview stats, ownership, the power/cost relationship, build-out timelines, and chip deployment volume.

Setup + load

```
In [1]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

sns.set_theme(style="whitegrid")
pd.set_option("display.max_columns", 20)

dc = pd.read_csv("../data/processed/data_centers_clean.csv")
dc.shape
```

Out[1]: (74, 18)

Overview

```
In [2]: dc.describe(include="all").T
```

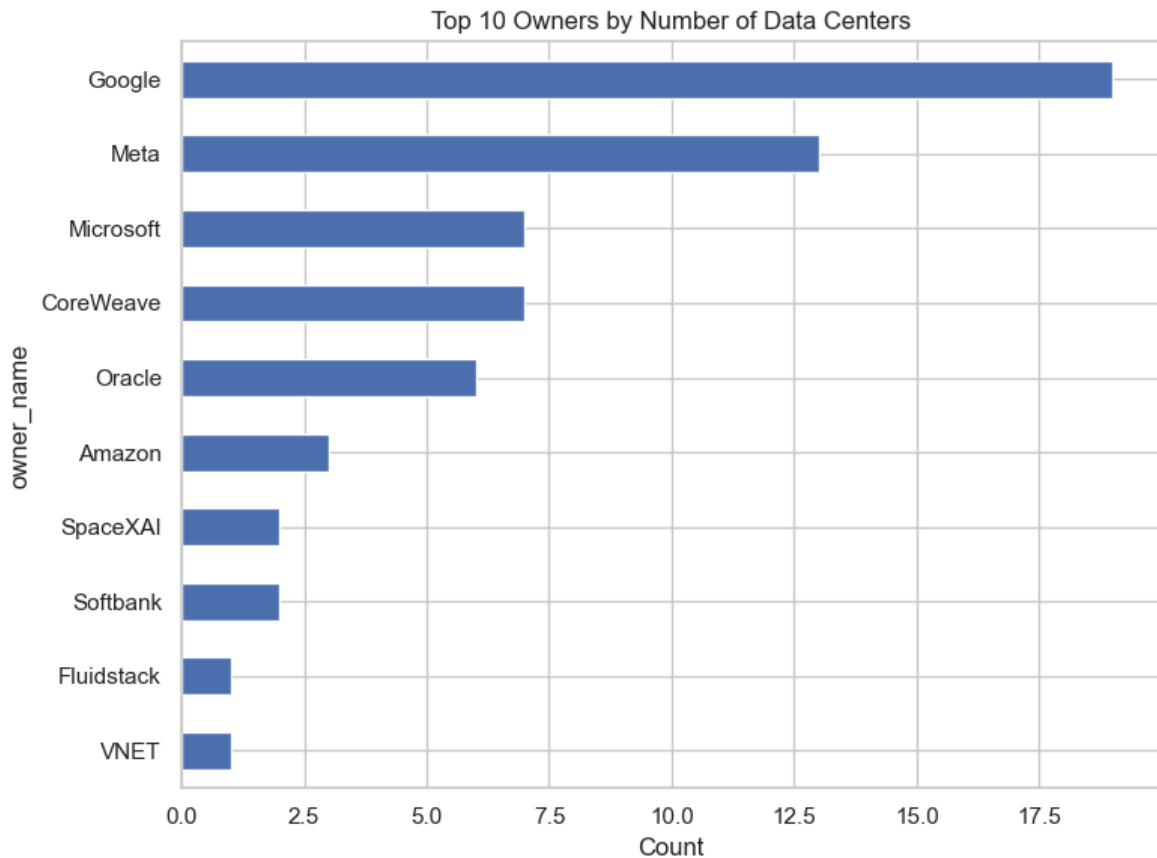
Out[2]:

	count	unique	
name	74	74	
current_h100_equivalents	74.0	NaN	
current_power_mw	74.0	NaN	
current_total_capital_cost_2025_usd_billions	74.0	NaN	
owner	66	17	
users	53	15	
selected_sources	74	74	- [WSJ] profile of the Colossus
calculations_sheet	74	74	https://docs.google.com/spreadsheets/d/1F8D9TtYUWzGgRQvXZlEjKqfVnHmLkPq/edit#gid=181111111
project	16	7	S&P 500
current_chip_types	50	22	
all_chip_types	60	25	
investors	7	5	
construction_companies	6	4	
energy_companies	15	12	
country	74	8	
address	64	63	5502 Spinks Rd
owner_name	66	15	
owner_confidence	66	3	

First question — biggest data centers by power

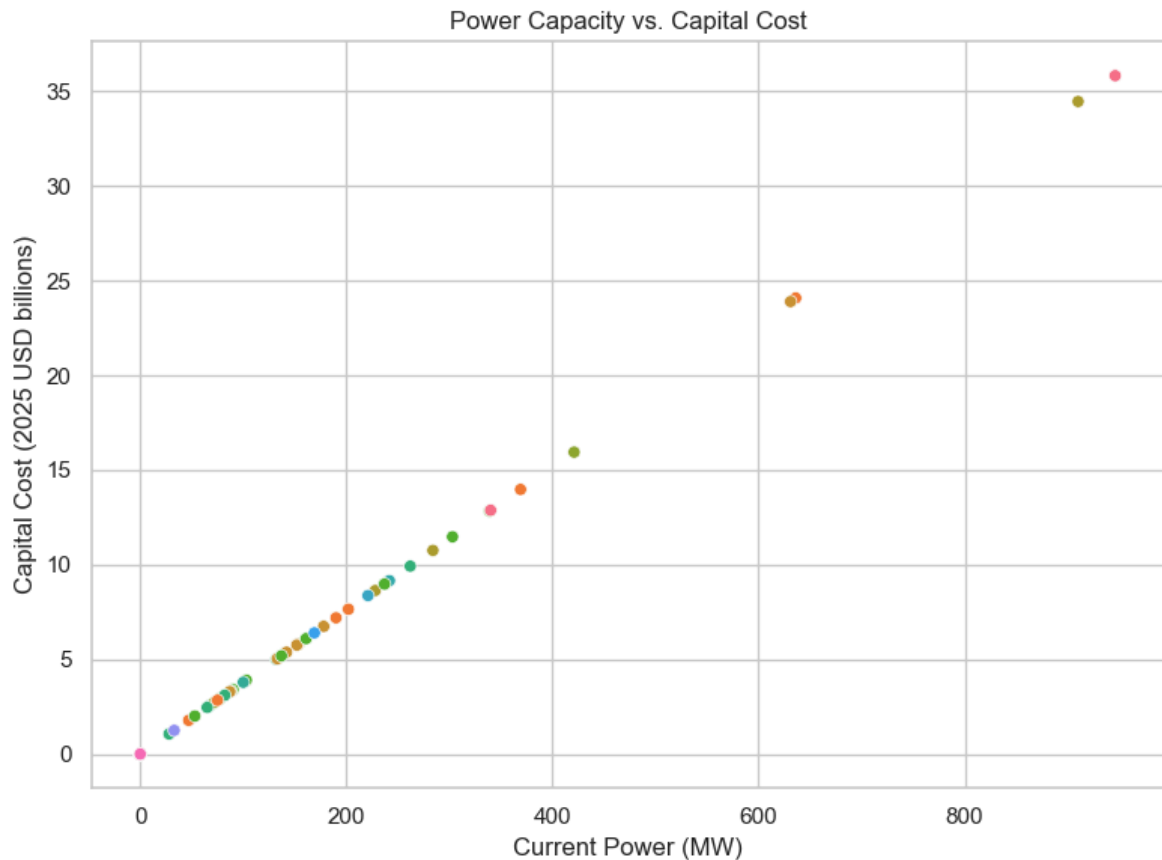
```
In [3]: dc = pd.read_csv("../data/processed/data_centers_clean.csv")

plt.figure(figsize=(8, 6))
dc["owner_name"].value_counts().head(10).plot(kind="barh")
plt.title("Top 10 Owners by Number of Data Centers")
plt.xlabel("Count")
plt.gca().invert_yaxis()
plt.tight_layout()
plt.show()
```



**Relationship between power and capital cost —
do bigger (more power) data centers cost
proportionally more?**

```
In [4]: plt.figure(figsize=(8, 6))
sns.scatterplot(data=dc, x="current_power_mw", y="current_total_capital_cost_2022",
                hue="owner_name", legend=False)
plt.title("Power Capacity vs. Capital Cost")
plt.xlabel("Current Power (MW)")
plt.ylabel("Capital Cost (2025 USD billions)")
plt.tight_layout()
plt.show()
```



```
In [5]: correlation = dc["current_power_mw"].corr(dc["current_total_capital_cost_2025_usd"])
print(f"Correlation: {correlation:.4f}")
```

Correlation: 1.0000

Finding: Power and Capital Cost are not independent

Correlation = 1.0000. Capital cost appears to be derived directly from power capacity (likely $\text{cost} = \text{power_mw} \times \text{constant}$), not independently reported. Treat these as one signal, not two, in any downstream analysis or dashboard — charting both separately would be redundant.

Load and inspect timelines

```
In [6]: timelines = pd.read_csv("../data/processed/data_center_timelines_clean.csv", parse_dates=True)
timelines["date"].min(), timelines["date"].max()
```

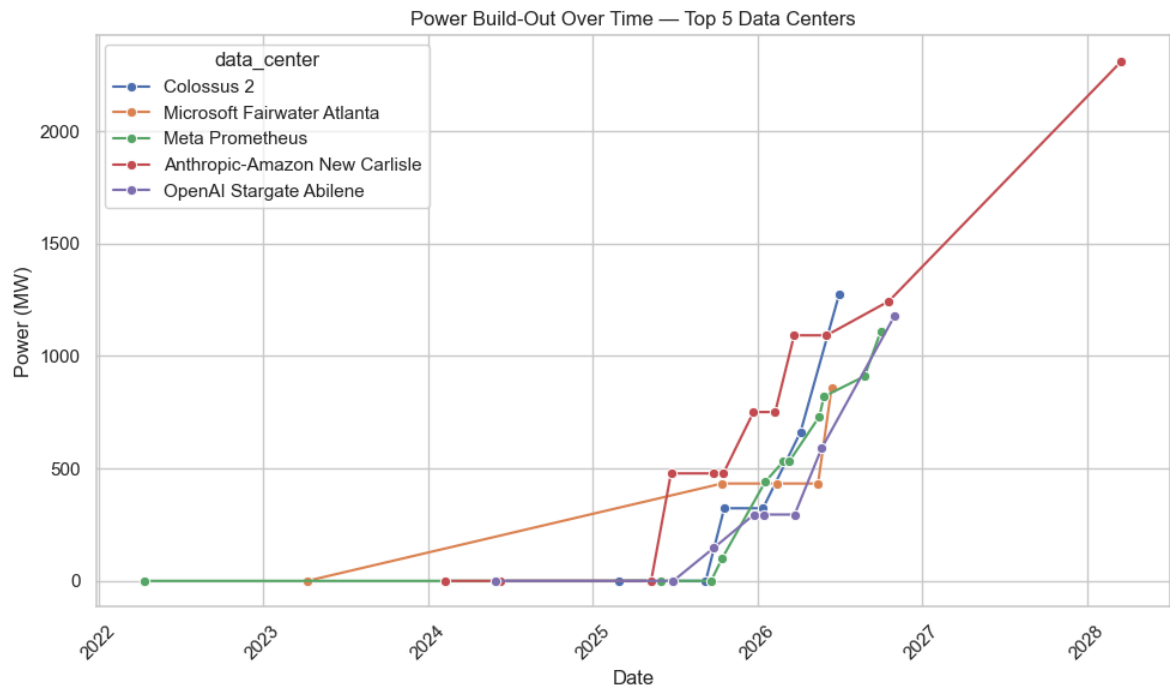
Out[6]: (Timestamp('2018-12-18 00:00:00'), Timestamp('2030-01-01 00:00:00'))

Power build-out for the top 5 data centers by current power

```
In [7]: top5_names = dc.sort_values("current_power_mw", ascending=False).head(5)["name"]
top5_timeline = timelines[timelines["data_center"].isin(top5_names)]

plt.figure(figsize=(10, 6))
sns.lineplot(data=top5_timeline, x="date", y="power_mw", hue="data_center", marker="o")
plt.title("Power Build-Out Over Time – Top 5 Data Centers")
plt.xlabel("Date")
plt.ylabel("Power (MW)")
```

```
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```



data_center_chip_quantities_clean.csv — it has actual unit counts per chip type per site, so we can see total deployed volume, not just which chip types are merely present somewhere.

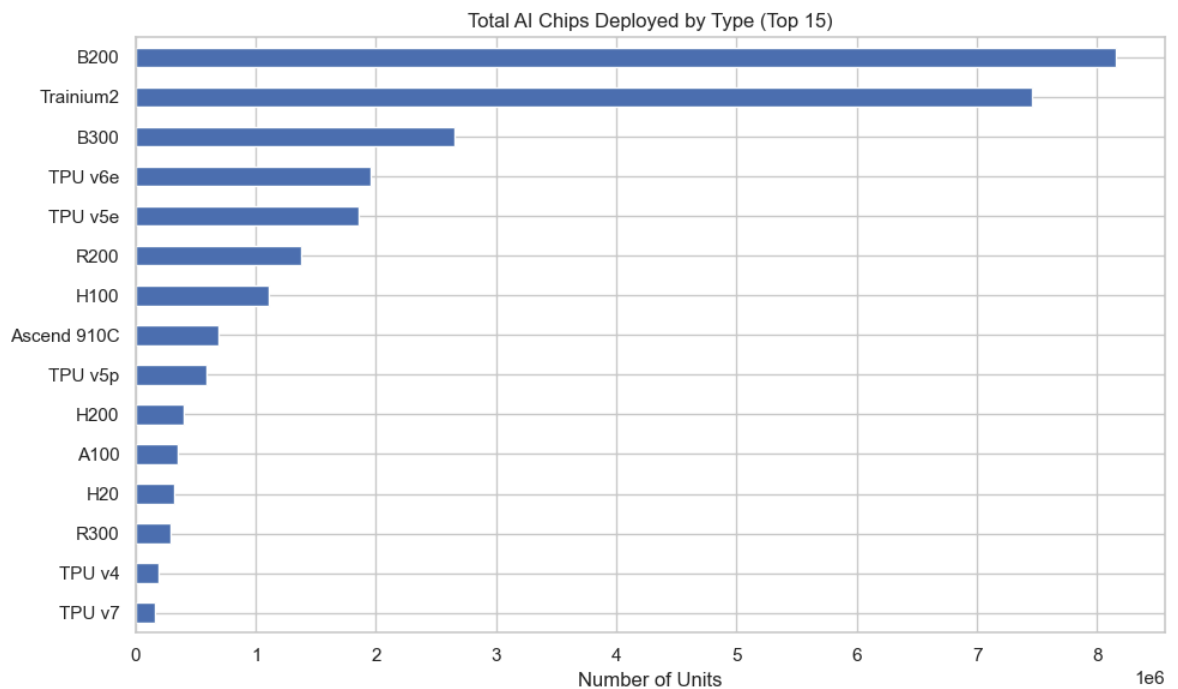
```
In [8]: chips = pd.read_csv("../data/processed/data_center_chip_quantities_clean.csv")

chip_totals = chips.groupby("chip_type")["number_of_units"].sum().sort_values(ascending=False)
chip_totals.head(15)
```

```
Out[8]: chip_type
B200          8155868.0
Trainium2     7460346.0
B300          2657329.0
TPU v6e       1959538.0
TPU v5e       1857320.0
R200          1380550.0
H100          1112866.0
Ascend 910C   685699.0
TPU v5p        592137.0
H200          402394.0
A100          353187.0
H20           319817.0
R300          291671.0
TPU v4         194429.0
TPU v7         163123.0
Name: number_of_units, dtype: float64
```

```
In [9]: plt.figure(figsize=(10, 6))
chip_totals.head(15).plot(kind="barh")
plt.title("Total AI Chips Deployed by Type (Top 15)")
plt.xlabel("Number of Units")
```

```
plt.ylabel("")
plt.gca().invert_yaxis()
plt.tight_layout()
plt.show()
```



Finding: Chip deployment is led by NVIDIA Blackwell, but custom silicon is major

B200 (NVIDIA Blackwell) leads at ~8.2M units, but Amazon's Trainium2 (~7.5M) and Google's TPU family are close behind — custom/in-house accelerators make up a large share of total deployed compute, not just NVIDIA GPUs.

PART 2 — Mining Previously-Unused Fields

Fields that existed in the source data from the start but were never analyzed: water use, energy efficiency by chip type, using the same 6 files as Part 1.

Water use — data_center_timelines_clean.csv has a water_use_mgd column (million gallons/day) that hasn't been touched yet

```
In [10]: water = timelines[timelines["water_use_mgd"].notna()]
print(f"{len(water)} of {len(timelines)} timeline rows report water use ({(water
water.sort_values("water_use_mgd", ascending=False)[["data_center", "date", "pow
```

67 of 419 timeline rows report water use (57 are exactly 0)

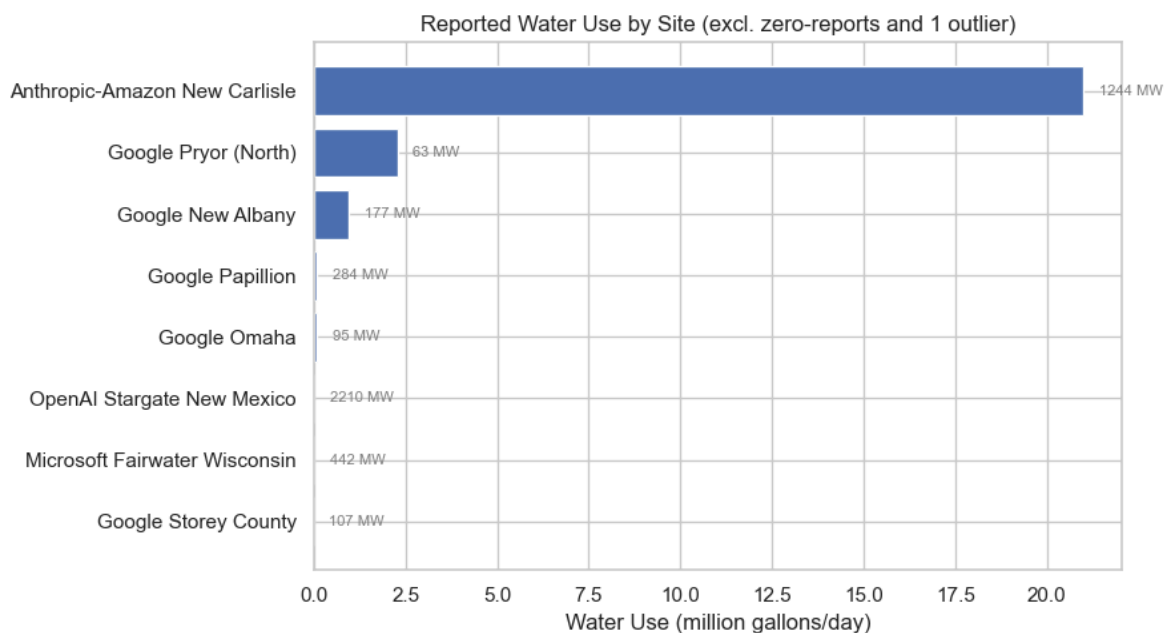
Out[10]:

	data_center	date	power_mw	water_use_mgd
167	Meta Kuna	2026-06-01	198.0	70000.00000
32	Anthropic-Amazon New Carlisle	2026-10-19	1244.0	21.00000
321	Google Pryor (North)	2024-07-19	63.0	2.28274
58	Google New Albany	2024-09-13	177.0	0.96600
57	Google New Albany	2024-04-20	81.0	0.84500

In [11]:

```
# Meta Kuna reports 70,000 MGD at 198 MW – implausible (that's more than NYC's e
# for a single 198 MW site) and swamps the axis. Treating it as a data error and
# A ranked bar avoids the labeling overlap a scatter+annotate produces with this
# Google New Albany reports twice (two build phases) – keep the latest so each s
water_nonzero = water[(water["water_use_mgd"] > 0) & (water["data_center"] != "M
water_nonzero = water_nonzero.sort_values("date").drop_duplicates("data_center",
water_bar = water_nonzero.sort_values("water_use_mgd")

plt.figure(figsize=(9, 5))
bars = plt.barh(water_bar["data_center"], water_bar["water_use_mgd"])
for bar, power in zip(bars, water_bar["power_mw"]):
    plt.text(bar.get_width() + water_bar["water_use_mgd"].max() * 0.02, bar.get_
             f"{power:.0f} MW", va="center", fontsize=8, color="gray")
plt.title("Reported Water Use by Site (excl. zero-reports and 1 outlier)")
plt.xlabel("Water Use (million gallons/day)")
plt.ylabel("")
plt.tight_layout()
plt.show()
```



Finding: water use is too sparsely reported to support a sustainability comparison

Only 67 of 419 timeline rows report `water_use_mgd` at all, 57 of those are exactly 0 (air/closed-loop cooled, or simply not disclosed — the data doesn't distinguish), leaving just 9 usable non-zero data points plus one clear data error (Meta Kuna, 70,000 MGD at 198 MW, excluded above). No visible relationship to power capacity at this sample size.

This field cannot support a "which data centers use less water" comparison as-is — it would need either more disclosure or a proxy (e.g. cooling system type from the chillers/cooling-towers tables) to be usable.

Energy efficiency by chip type — gpu_clusters_clean.csv, unused so far

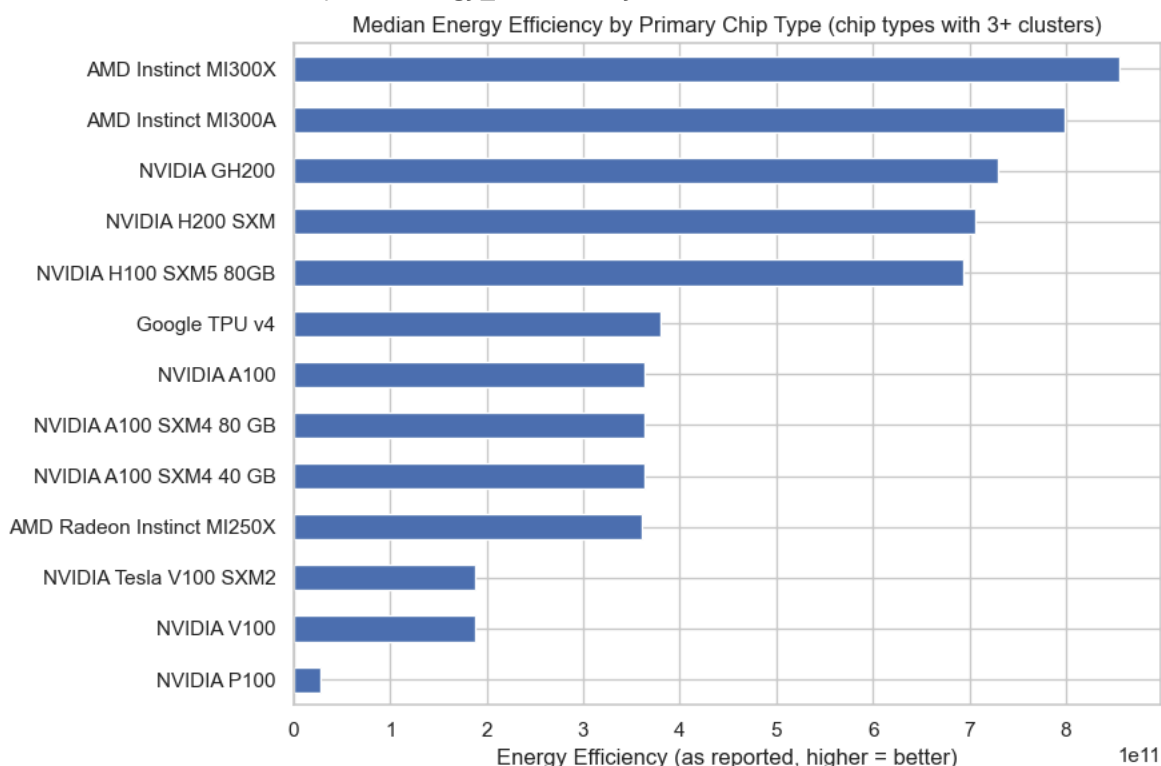
This file is a broader table of GPU clusters (482 rows) than `data_centers_clean.csv` (74 rows), and includes an `energy_efficiency` column plus per-site `latitude` / `longitude`.

```
In [12]: gpu = pd.read_csv("../data/processed/gpu_clusters_clean.csv")
print(f"{gpu['energy_efficiency'].notna().sum()} of {len(gpu)} clusters report e

eff_by_chip = (
    gpu[gpu["energy_efficiency"].notna()]
    .groupby("chip_type_primary")["energy_efficiency"]
    .agg(["median", "count"])
    .query("count >= 3")
    .sort_values("median", ascending=False)
)

plt.figure(figsize=(9, 6))
eff_by_chip["median"].plot(kind="barh")
plt.title("Median Energy Efficiency by Primary Chip Type (chip types with 3+ clu
plt.xlabel("Energy Efficiency (as reported, higher = better)")
plt.ylabel("")
plt.gca().invert_yaxis()
plt.tight_layout()
plt.show()
```

406 of 482 clusters report energy_efficiency



Finding: newer chip generations rank most energy-efficient, but the source doesn't state units

AMD MI300X/MI300A, NVIDIA GH200, and H200 top the ranking; NVIDIA A100 (the oldest chip with meaningful sample size, 73-81 clusters) ranks lowest. That tracks with known hardware trends — each generation ships more compute per watt — but the `energy_efficiency` column in the source data has no documented unit, so treat this as a *relative* ranking (newer > older) rather than a comparable absolute metric across chip types.

PART 3 — External Sustainability Data & the 10 Analytical Questions

From here on, this notebook pulls in an external dataset (OWID electricity data, cleaned in `clean_data_centers.ipynb`) to work through the 10 analytical questions from the project brief — country-level renewable share, carbon intensity, clustering, operator comparisons, and geographic concentration.

Country sustainability & cost profile — joining in external data

None of the 6 source files contain electricity price, carbon emissions, or renewable-energy-% data — a real gap for the "where should AI data centers be built sustainably?" question this notebook works through below. Two external datasets close that gap, both cleaned in `clean_data_centers.ipynb`:

- **Sustainability:** [OWID's energy dataset](#) → `data/processed/owid_country_sustainability_clean.csv` (renewable share, carbon intensity, latest year per country)
- **Cost:** a manually compiled snapshot from [globalpetrolprices.com](#) → `data/processed/electricity_price_clean.csv` (residential USD/kWh, 2023-2026 average — not a live feed, see caveat in the cleaning notebook)

Both join onto `gpu_clusters_clean.csv` by country.

```
In [13]: sustainability = pd.read_csv("../data/processed/owid_country_sustainability_clean.csv")
prices = pd.read_csv("../data/processed/electricity_price_clean.csv")

country_sustainability = (
    gpu.merge(sustainability[["country_epoch", "renewables_share_elec", "carbon_intensity_elec"],
                           left_on="country", right_on="country_epoch", how="left"])
    .merge(prices[["country_epoch", "price_usd_per_kwh"]], left_on="country", right_on="country_epoch", how="left")
    .groupby("country")
    .agg(
        clusters=("name", "count"),
        total_power_mw=("power_capacity_mw", "sum"),
```

```
renewables_share_elec=("renewables_share_elec", "first"),
carbon_intensity_elec=("carbon_intensity_elec", "first"),
price_usd_per_kwh=("price_usd_per_kwh", "first"),
)
.query("clusters >= 3")
.sort_values("clusters", ascending=False)
)
country_sustainability
```

Out[13]:

	clusters	total_power_mw	renewables_share_elec	carbon_intensity_elec	pri
--	----------	----------------	-----------------------	-----------------------	-----

country					
China	188	296.900000	37.044	525.34	
United States of America	119	1962.754570	25.638	384.40	
Japan	25	43.102966	23.569	477.26	
France	14	33.175048	26.064	41.44	
Korea (Republic of)	12	12.735673	10.394	417.06	
Germany	11	25.171676	59.092	329.65	
Italy	9	39.944208	48.776	284.78	
Brazil	9	7.265970	86.601	109.95	
Russia	8	5.720723	17.353	449.73	
United Kingdom of Great Britain and Northern Ireland	6	11.996963	51.972	217.41	
Thailand	5	3.464350	16.575	545.74	
Canada	5	5.507334	63.900	190.72	
Switzerland	4	26.777692	67.825	39.22	
Sweden	4	7.432123	71.191	35.26	
Finland	4	25.966398	56.464	57.47	
Poland	4	1.787102	31.508	588.60	
India	3	6.682458	24.067	670.13	
Australia	3	1.890870	38.618	525.18	
Netherlands	3	2.111549	51.200	253.56	
Norway	3	29.222502	98.999	28.11	
Saudi Arabia	3	14.181357	2.160	691.95	
Singapore	3	1.978995	5.486	497.09	
Taiwan	3	4.169312	12.240	633.21	
United Arab Emirates	3	3.040491	8.812	467.51	

Q1 — Which countries host the largest concentration of AI-capable data centers? (Choropleth map)

Fills each country polygon by its GPU cluster count. Joins on ISO-3 code (`iso_code` from the OWID file / `ISO_A3` in the Natural Earth boundaries) rather than country name, since it's the more reliable key.

```
In [14]: import json
import matplotlib.colors as mcolors

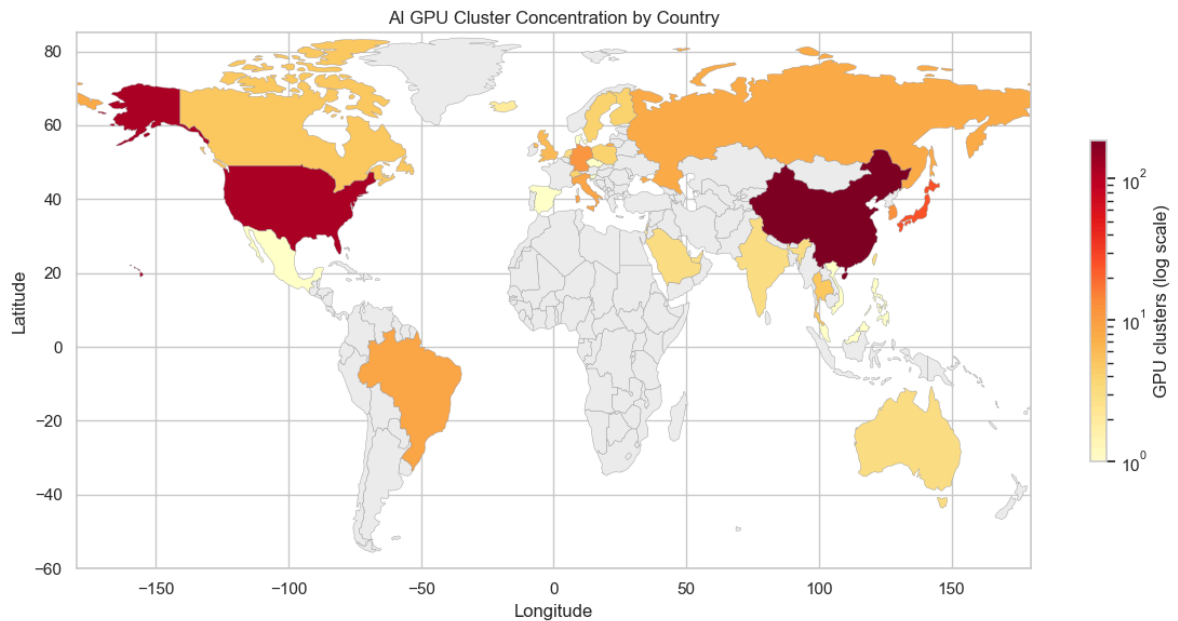
with open("../data/raw/ne_110m_countries.geojson", encoding="utf-8") as f:
    world = json.load(f)

country_counts = gpu.groupby("country").size().reset_index(name="clusters")
country_counts = country_counts.merge(sustainability[["country_epoch", "iso_code"],
                                                    left_on="country", right_on="country_epoch"],
                                     how="left")
count_by_iso = country_counts.set_index("iso_code")["clusters"]

cmap = plt.colormaps["YlOrRd"]
norm = mcolors.LogNorm(vmin=1, vmax=count_by_iso.max())

fig, ax = plt.subplots(figsize=(12, 6))
for feature in world["features"]:
    iso = feature["properties"].get("ISO_A3")
    count = count_by_iso.get(iso)
    color = cmap(norm(count)) if count and count > 0 else "#eeeeee"
    geom = feature["geometry"]
    polys = geom["coordinates"] if geom["type"] == "MultiPolygon" else [geom["coordinates"]]
    for poly in polys:
        xs, ys = zip(*poly[0])
        ax.fill(xs, ys, facecolor=color, edgecolor="#999999", linewidth=0.3)

sm = plt.cm.ScalarMappable(cmap=cmap, norm=norm)
sm.set_array([])
plt.colorbar(sm, ax=ax, label="GPU clusters (log scale)", shrink=0.6)
ax.set_xlim(-180, 180)
ax.set_ylim(-60, 85)
ax.set_title("AI GPU Cluster Concentration by Country")
ax.set_xlabel("Longitude")
ax.set_ylabel("Latitude")
plt.tight_layout()
plt.show()
```



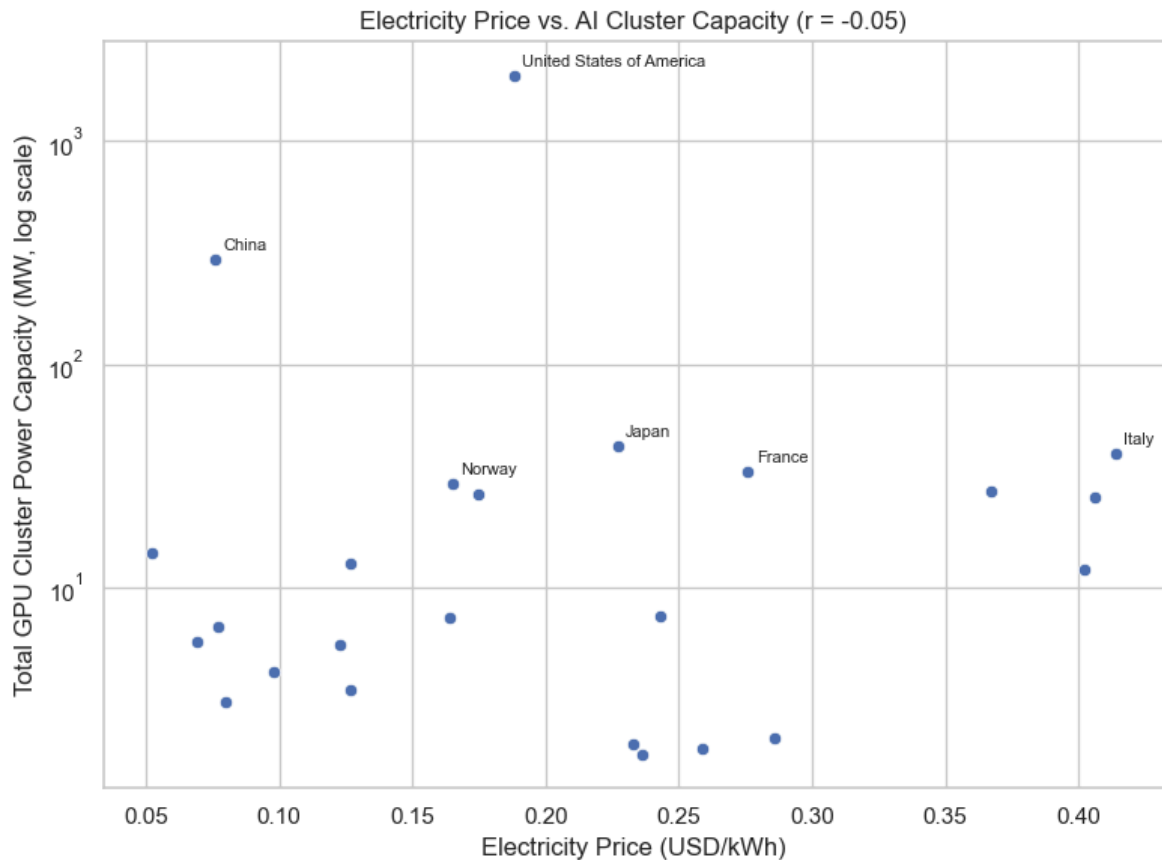
Finding: China leads on cluster count, not just the US

By cluster count, China (188) actually outnumbers the United States (119) in this table, followed by Japan (25), France (14), and Germany (11) — a very different picture than the ownership breakdown from `data_centers_clean.csv` earlier in this notebook, which was almost entirely US hyperscalers. That table is Epoch AI's curated set of the largest/most notable AI data centers; `gpu_clusters_clean.csv` is a broader survey, which is why it's the one used for this map.

Q2 — Does cheaper electricity correlate with more data center capacity? (Scatter plot)

Using `price_usd_per_kwh` against each country's total GPU cluster power capacity (log scale — the US and China dominate so heavily that a linear axis hides everyone else).

```
In [15]: plt.figure(figsize=(8, 6))
sns.scatterplot(data=country_sustainability, x="price_usd_per_kwh", y="total_power_mw")
plt.yscale("log")
for country in country_sustainability.nlargest(6, "total_power_mw").index:
    row = country_sustainability.loc[country]
    plt.annotate(country, (row["price_usd_per_kwh"], row["total_power_mw"]),
                  fontsize=8, xytext=(4, 4), textcoords="offset points")
r = country_sustainability["price_usd_per_kwh"].corr(country_sustainability["total_power_mw"])
plt.title(f"Electricity Price vs. AI Cluster Capacity (r = {r:.2f})")
plt.xlabel("Electricity Price (USD/kWh)")
plt.ylabel("Total GPU Cluster Power Capacity (MW, log scale)")
plt.tight_layout()
plt.show()
```



Finding: no — cheaper electricity does not predict AI capacity

$r = -0.05$, essentially no linear relationship. China (0.076 USD/kWh) does have massive capacity, but the US (0.188 USD/kWh — pricier than Russia, India, Taiwan, Saudi Arabia, and South Korea, all far smaller AI hosts) still has 6-7x China's capacity. Whatever actually drives where AI infrastructure concentrates, it isn't raw electricity price — capital availability, existing cloud/tech ecosystems, chip supply access, and policy plausibly matter far more.

Q3 — Which operators have the highest renewable energy adoption? (Bar chart)

Important caveat: we don't have each operator's actual clean-energy procurement (many hyperscalers buy PPAs/RECs that make their real usage cleaner than the local grid). This is a proxy — each operator's `owner_name` sites in `data_centers_clean.csv`, weighted by `current_power_mw`, mapped to the **grid's** renewable share in the country each site sits in. It measures "how clean is the grid where this operator builds," not "how clean is this operator."

```
In [16]: dc_renew = dc.merge(sustainability[["country", "renewables_share_elec"]], on="co

def weighted_or_mean(g):
    weights = g["current_power_mw"]
    if weights.sum() == 0:
        return g["renewables_share_elec"].mean()
```

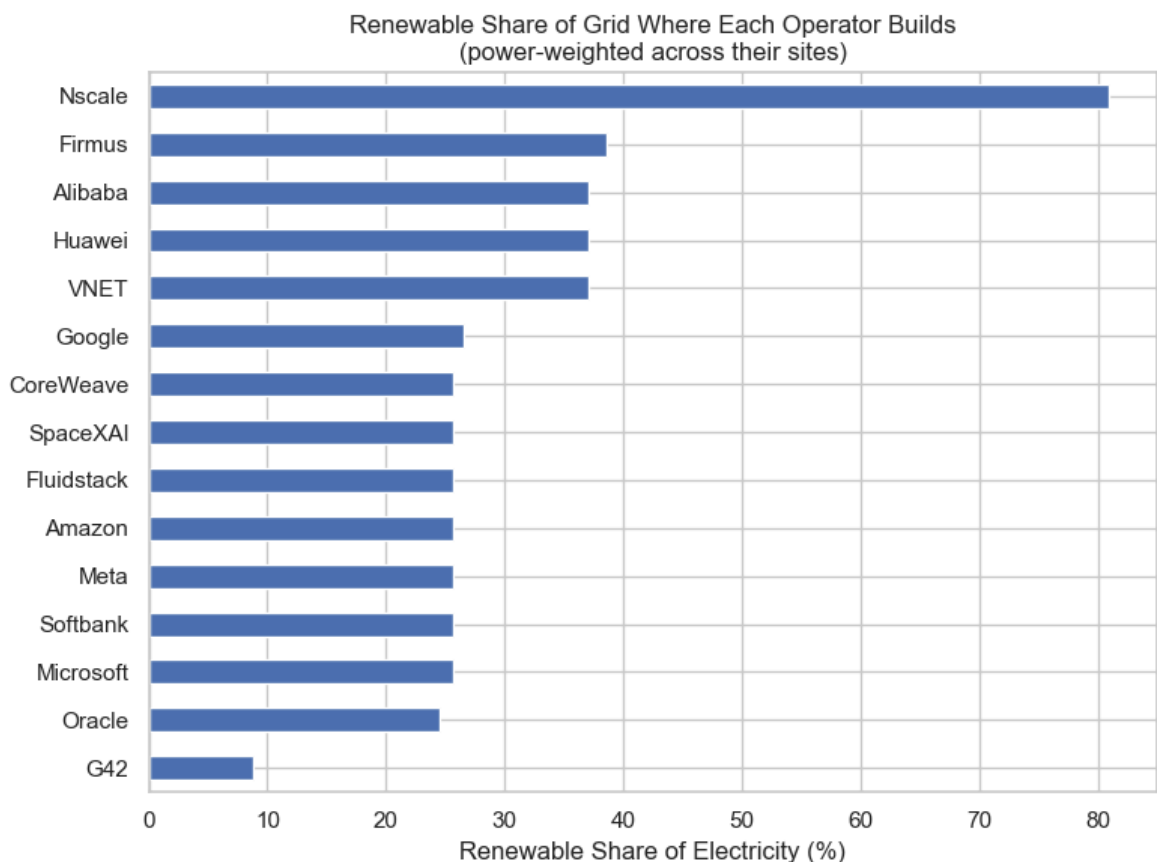
```

return np.average(g["renewables_share_elec"], weights=weights)

operator_renew = (
    dc_renew.groupby("owner_name")
    .apply(weighted_or_mean, include_groups=False)
    .rename("renewable_share_weighted")
    .sort_values(ascending=False)
)

plt.figure(figsize=(8, 6))
operator_renew.plot(kind="barh")
plt.title("Renewable Share of Grid Where Each Operator Builds\n(power-weighted across their sites)")
plt.xlabel("Renewable Share of Electricity (%)")
plt.ylabel("")
plt.gca().invert_yaxis()
plt.tight_layout()
plt.show()

```



Finding: most major hyperscalers land in a tight band — a few single-site outliers stand apart

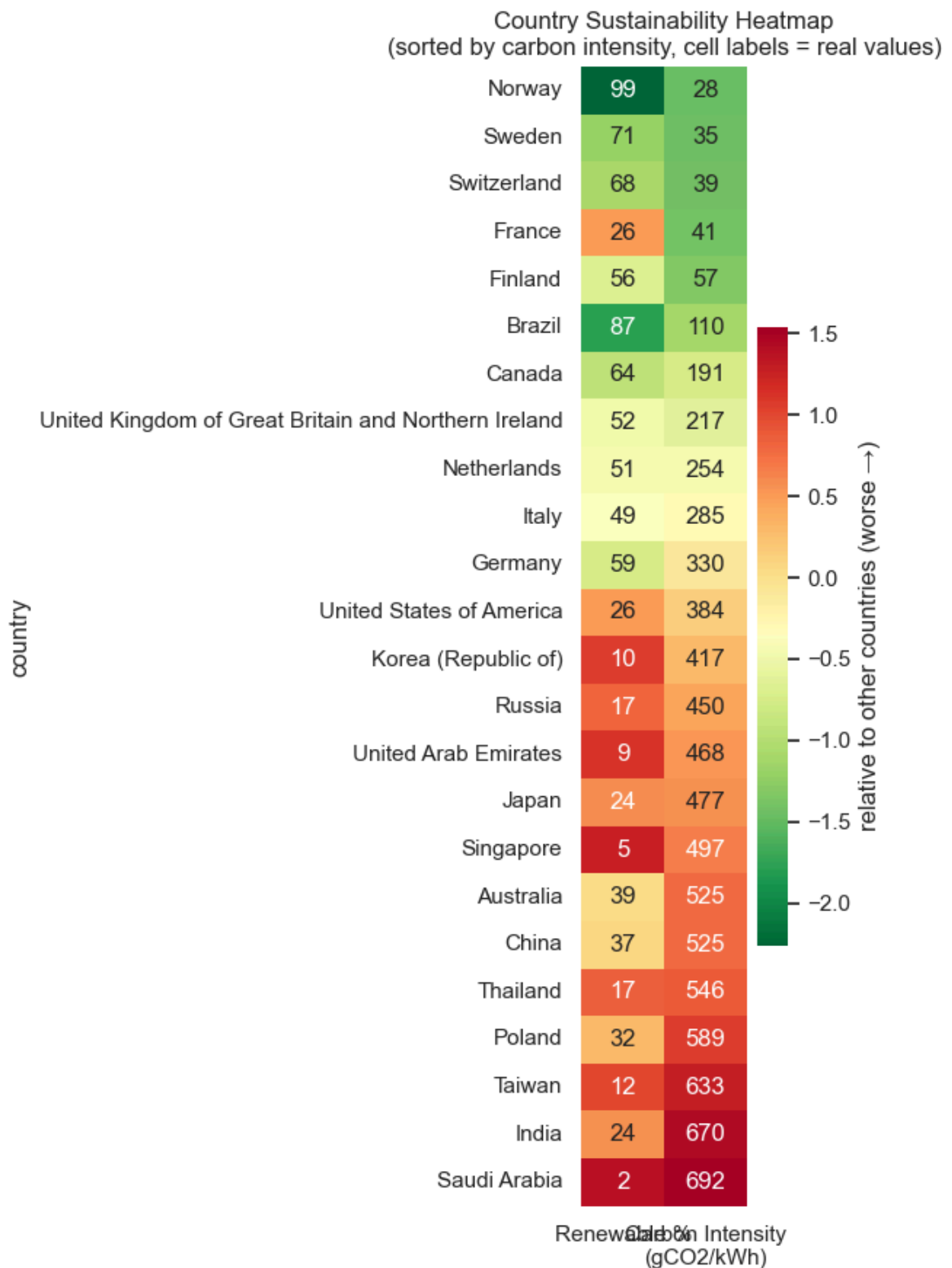
Nscale (~81%) is the clear outlier at the clean end (single European site), and G42 (~9%) at the dirty end. But the big names — Google, CoreWeave, SpaceXAI, Fluidstack, Amazon, Meta, Softbank, Microsoft, Oracle — all cluster tightly around 24-26%, because they're overwhelmingly US-sited and that's just the US grid average. This is grid cleanliness, not operator behavior — see the caveat above. A single-site operator's number says more about which one country they happen to be in than about any real efficiency difference.

Q4 — How does carbon intensity differ across countries with major AI infrastructure? (Heatmap)

```
In [17]: heat_data = country_sustainability[["renewables_share_elec", "carbon_intensity_elec"]]

# Z-score each column so both share one color scale. Negate renewables_share_elec
# so the colormap direction is consistent: red = bad (high carbon / low renewables)
heat_color = (heat_data - heat_data.mean()) / heat_data.std()
heat_color["renewables_share_elec"] *= -1

plt.figure(figsize=(6, 9))
sns.heatmap(heat_color, annot=heat_data.round(0), fmt=".0f", cmap="RdYlGn_r",
            cbar_kws={"label": "relative to other countries (worse →)"},
            xticklabels=["Renewable %", "Carbon Intensity\n(gCO2/kWh)"])
plt.title("Country Sustainability Heatmap\n(sorted by carbon intensity, cell labels)")
plt.tight_layout()
plt.show()
```

Finding: a 24x spread between the cleanest and dirtiest grids in this set

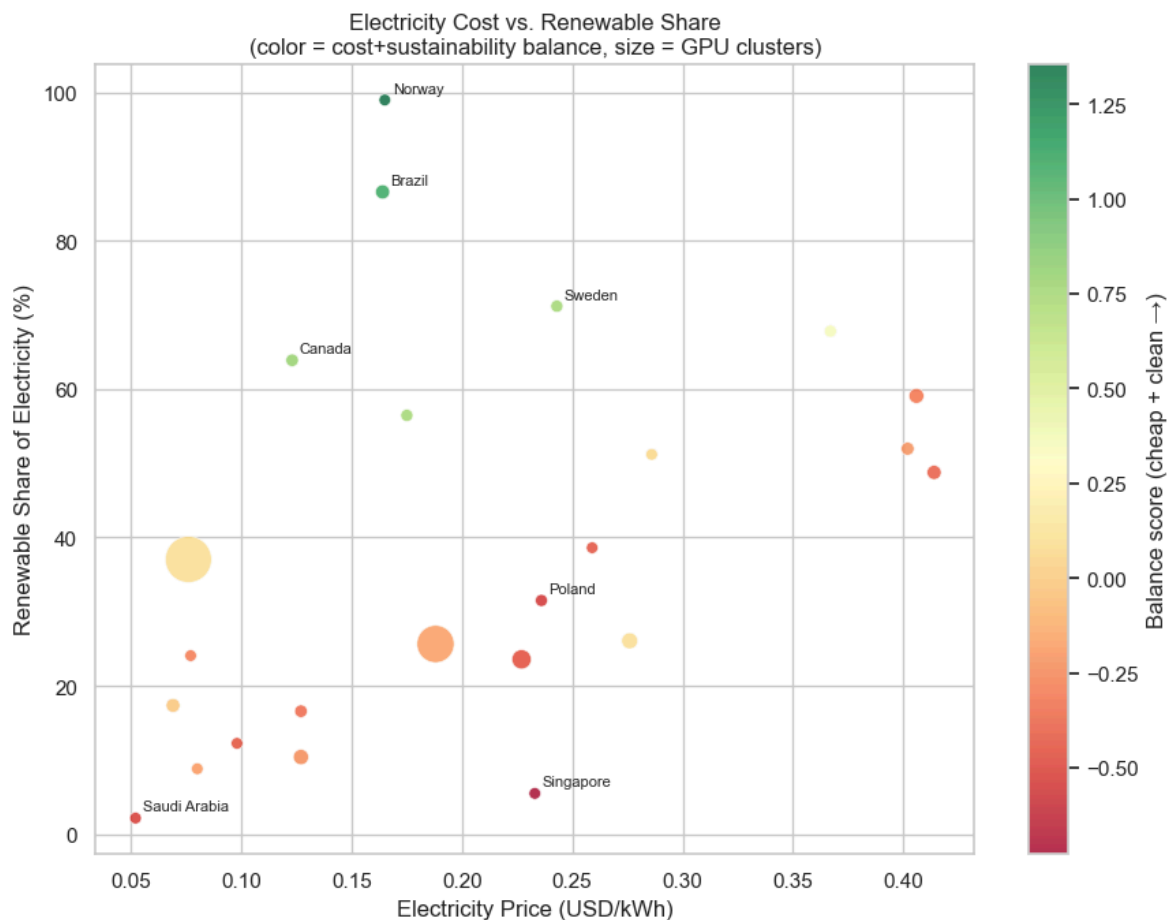
Norway sits at 28 gCO₂/kWh; Saudi Arabia at 692 gCO₂/kWh — both host meaningful AI infrastructure today. Countries hosting AI clusters span nearly the entire global range of grid cleanliness; there's no cluster of infrastructure only in clean grids or only in dirty ones.

Q5 — Which countries provide the best balance of electricity cost and sustainability? (Bubble chart)

Composite of 3 z-scored signals — cheap electricity, high renewable share, low carbon intensity — averaged into one score. Bubble size = number of GPU clusters (today's activity level, shown for reference; the score itself isn't weighted by it).

```
In [18]: z = lambda s: (s - s.mean()) / s.std()
country_sustainability["balance_score"] = (
    -z(country_sustainability["price_usd_per_kwh"])
    + z(country_sustainability["renewables_share_elec"])
    - z(country_sustainability["carbon_intensity_elec"])
) / 3

plt.figure(figsize=(9, 7))
sc = plt.scatter(country_sustainability["price_usd_per_kwh"], country_sustainabi
                  s=country_sustainability["clusters"] * 3 + 30, c=country_susta
                  cmap="RdYlGn", alpha=0.8, edgecolor="white", linewidth=0.5)
standouts = pd.concat([country_sustainability["balance_score"].nlargest(4),
                       country_sustainability["balance_score"].nsmallest(3)].in
for country in standouts:
    row = country_sustainability.loc[country]
    plt.annotate(country, (row["price_usd_per_kwh"], row["renewables_share_elec"]
                       fontsize=8, xytext=(4, 4), textcoords="offset points")
plt.colorbar(sc, label="Balance score (cheap + clean →)")
plt.title("Electricity Cost vs. Renewable Share\n(color = cost+sustainability ba
plt.xlabel("Electricity Price (USD/kWh)")
plt.ylabel("Renewable Share of Electricity (%)")
plt.tight_layout()
plt.show()
```



Finding: Norway and Brazil lead; China's cost advantage changes its ranking

Norway (cheap + 99% renewable) and Brazil (cheap + 87% renewable) top the composite. China is notably higher here than in the sustainability-only view (Q4/Q8/Q10) because it's genuinely cheap (0.076 USD/kWh) even though its grid isn't especially clean — cost and sustainability don't reward the same countries. The US ranks below average on this composite (mid-pack sustainability, above-average price) despite hosting the most capacity by far — today's build-out pattern doesn't track this balance score.

Q6 — Where are new AI data centers being built most rapidly? (Timeline)

Uses `data_center_timelines_clean.csv` — each data center's earliest tracked date as its "project start," grouped by year and country. Note this table only covers the 74 sites in `data_centers_clean.csv` (Epoch's curated large/notable set), which is heavily US-weighted — treat the country split as directional, not representative of global build activity the way `gpu_clusters_clean.csv` is elsewhere in this notebook.

```
In [19]: first_seen = (
    timelines.merge(dc[["name", "country"]], left_on="data_center", right_on="name")
    .groupby("data_center")
    .agg(country=("country", "first"), start_date=("date", "min"))
)
first_seen["start_year"] = first_seen["start_date"].dt.year
```

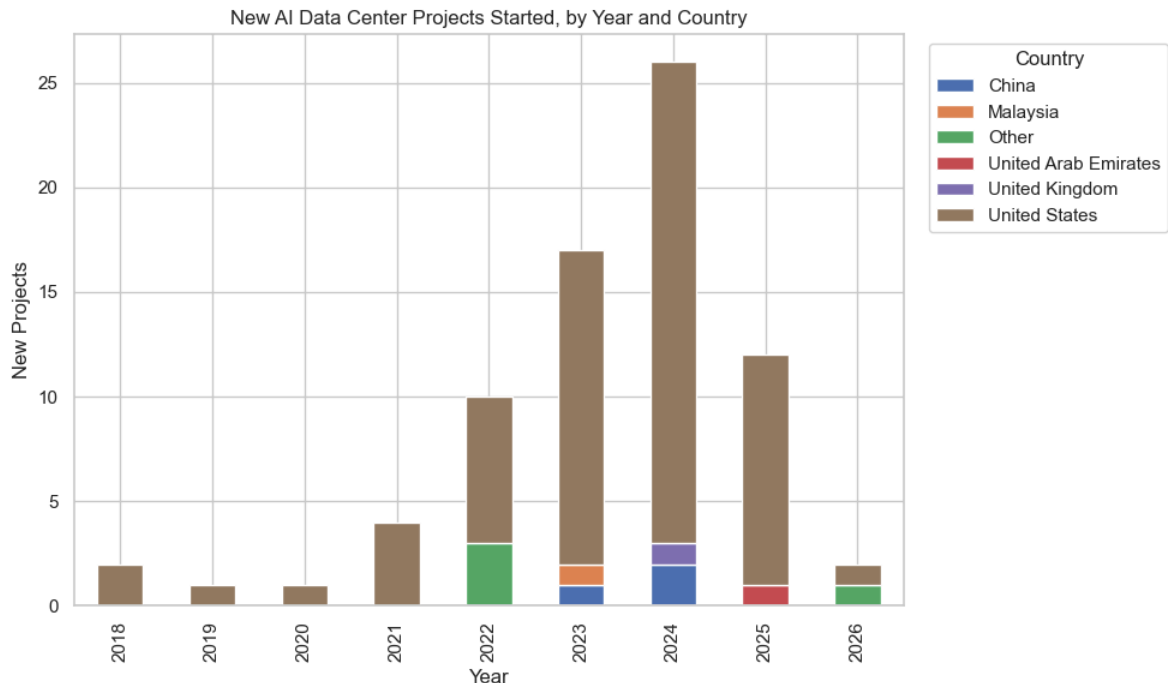
```

top_countries = first_seen["country"].value_counts().head(5).index
first_seen["country_grouped"] = first_seen["country"].where(first_seen["country"]

by_year_country = first_seen.groupby(["start_year", "country_grouped"]).size().u

by_year_country.plot(kind="bar", stacked=True, figsize=(10, 6))
plt.title("New AI Data Center Projects Started, by Year and Country")
plt.xlabel("Year")
plt.ylabel("New Projects")
plt.legend(title="Country", bbox_to_anchor=(1.02, 1), loc="upper left")
plt.tight_layout()
plt.show()

```



Finding: 2024 was the peak year, and it's overwhelmingly a US story

26 new tracked projects started in 2024 (up from 10 in 2022, 17 in 2023) before dropping to 12 in 2025 — more than half of all 74 tracked projects started in 2023-2024 alone, and the United States accounts for the large majority of every year's total. Remember the caveat above: this table only covers Epoch's curated 74-site set, not the broader global picture from `gpu_clusters_clean.csv`.

Q7 — How does renewable energy usage relate to estimated emissions? (Scatter plot with regression)

Using `carbon_intensity_elec` as the emissions measure (grams CO2 per kWh actually generated, rather than a country total, which would just track population/economy size instead of grid cleanliness).

```

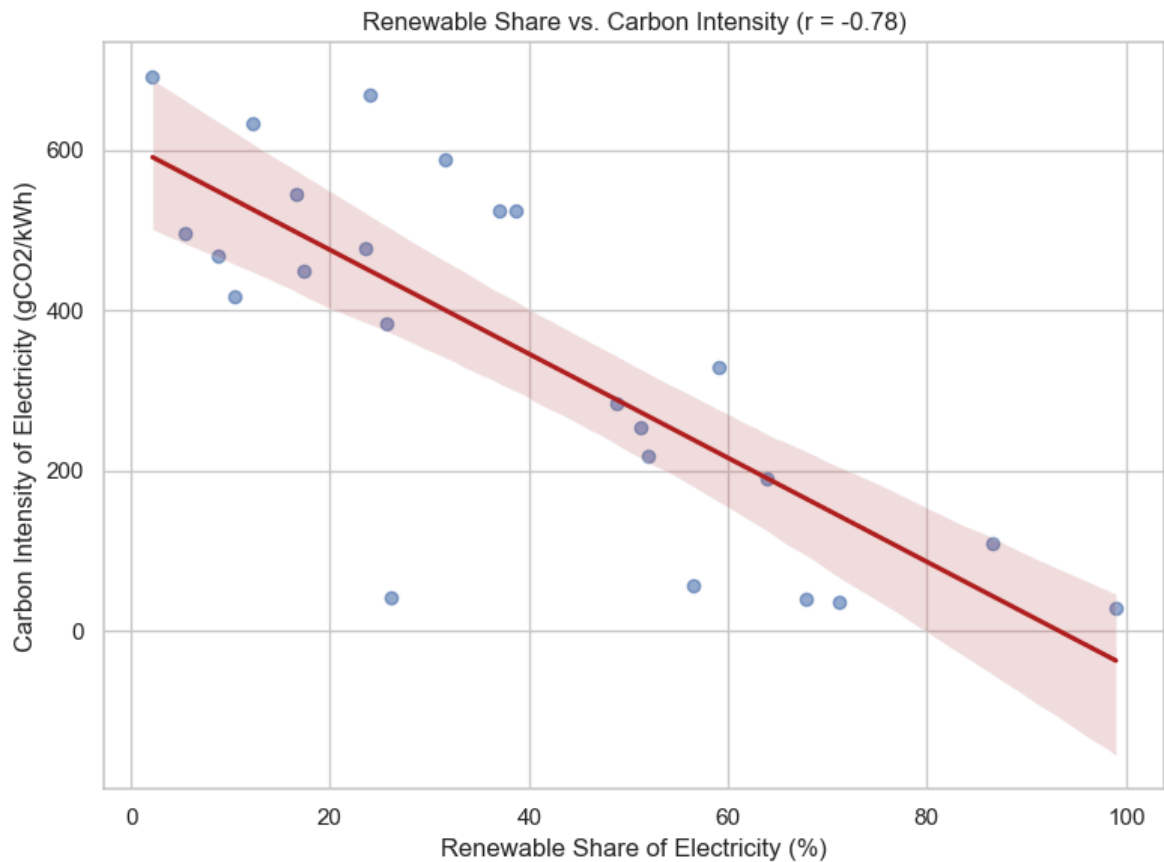
In [20]: plt.figure(figsize=(8, 6))
sns.regplot(data=country_sustainability, x="renewables_share_elec", y="carbon_in

```

```

scatter_kws={"alpha": 0.6}, line_kws={"color": "firebrick"})
r = country_sustainability["renewables_share_elec"].corr(country_sustainability[
plt.title(f"Renewable Share vs. Carbon Intensity (r = {r:.2f})")
plt.xlabel("Renewable Share of Electricity (%)")
plt.ylabel("Carbon Intensity of Electricity (gCO2/kWh)")
plt.tight_layout()
plt.show()

```



Finding: a strong but imperfect relationship (r = -0.78)

Renewable share explains most of the variation in carbon intensity, but France is the clear exception, sitting well below the regression line — a nuclear-heavy grid reaches low emissions without high renewable share. Renewable % alone is a good but incomplete proxy for "clean grid," consistent with the Q4/K-Means findings above.

Q8 — Which regions are best suited for future AI expansion? (Ranking)

A composite score isn't in the source data — built here as the average of two z-scored signals: renewable share (higher better) and carbon intensity (lower better). This says nothing about land, grid capacity headroom, or policy — just "how clean is this country's electricity today."

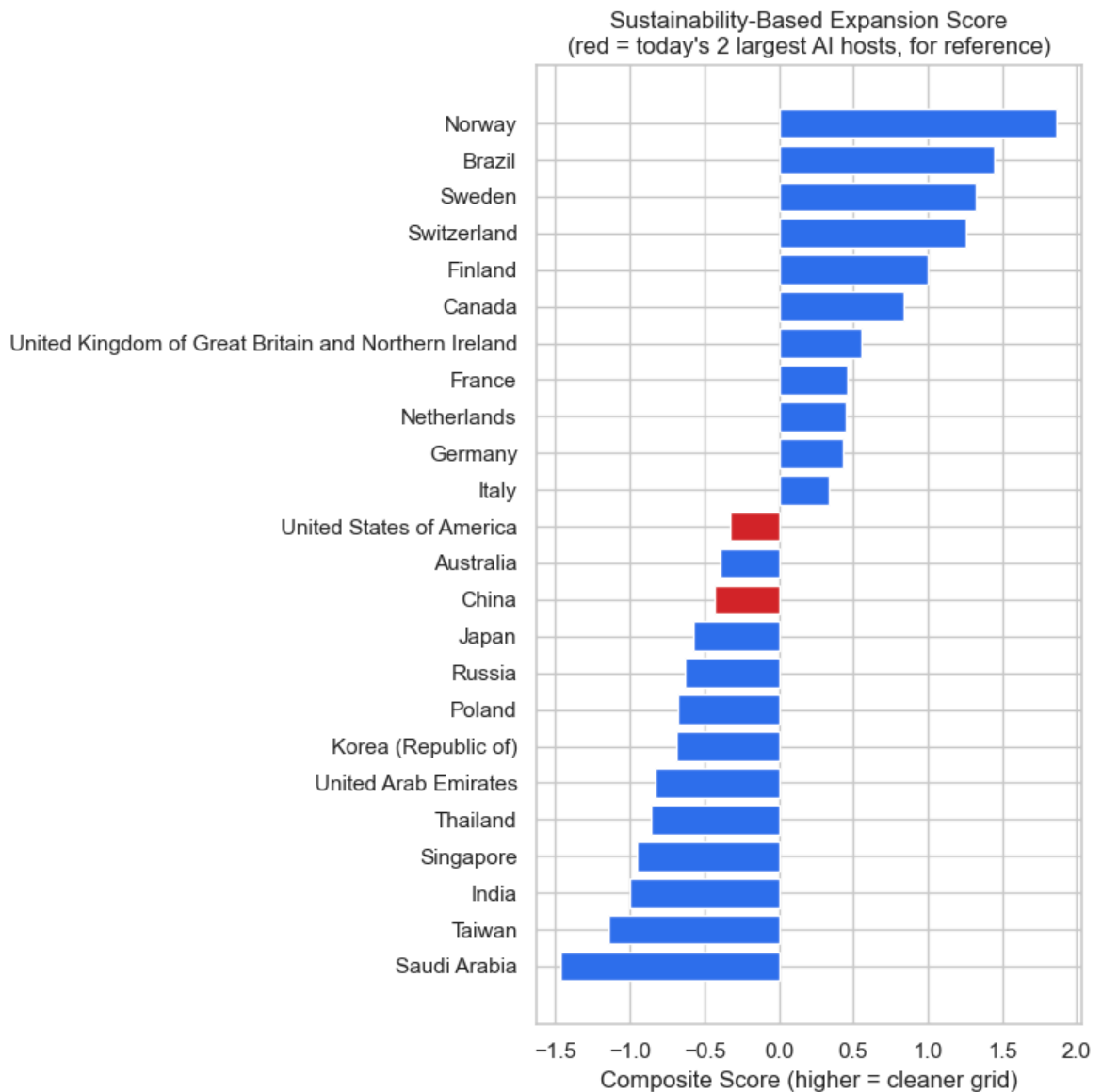
```

In [21]: z_renew = (country_sustainability["renewables_share_elec"] - country_sustainabil
z_carbon = (country_sustainability["carbon_intensity_elec"] - country_sustainabi
expansion_score = ((z_renew - z_carbon) / 2).sort_values(ascending=False)

colors = ["#d62728" if c in ("United States of America", "China") else "#2f6fed"

```

```
plt.figure(figsize=(8, 8))
plt.barh(expansion_score.index, expansion_score.values, color=colors)
plt.title("Sustainability-Based Expansion Score\n(red = today's 2 largest AI hos")
plt.xlabel("Composite Score (higher = cleaner grid)")
plt.gca().invert_yaxis()
plt.tight_layout()
plt.show()
```



Finding: Norway, Brazil, and Sweden top a sustainability-only ranking; Saudi Arabia bottom

This ranking ignores cost entirely (see Q5 for the cost-adjusted version, which reorders things — China rises once cost is factored in). Both today's largest hosts, the US and China, sit in the lower half on sustainability alone.

Q9 — How do cloud providers compare in global infrastructure footprint? (Treemap)

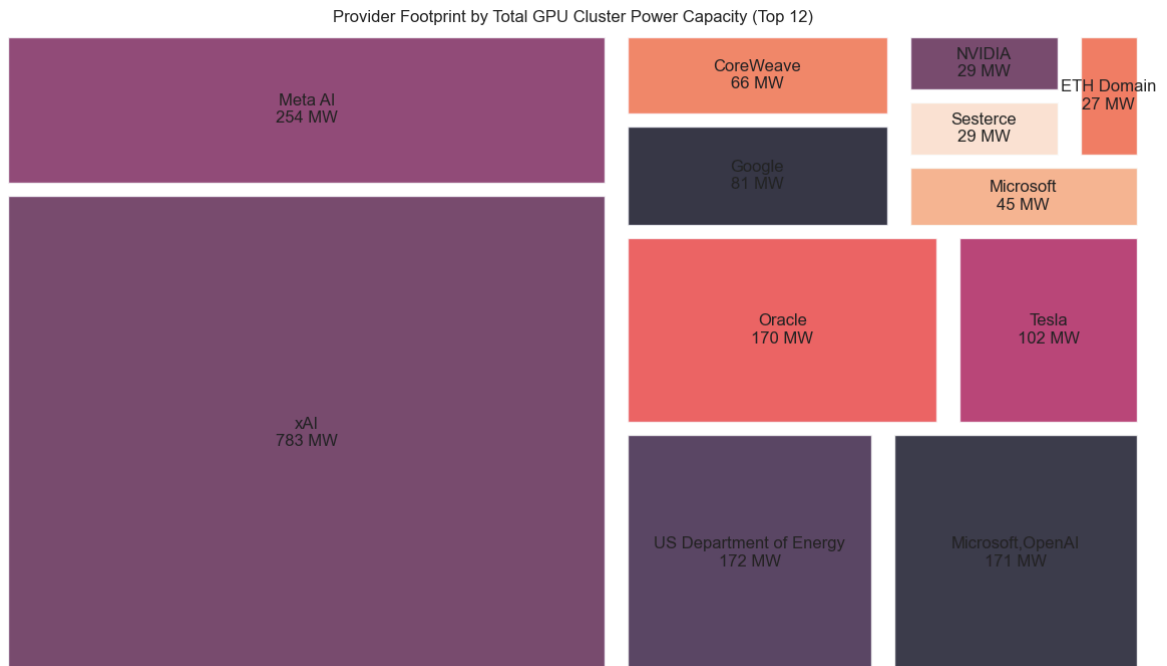
Using `gpu_clusters_clean.csv`'s `owner` field (broader coverage than `data_centers_clean.csv`), summed power capacity by owner. 198 of 482 clusters

have no reported owner and are excluded.

```
In [22]: import squarify

owner_power = gpu.dropna(subset=["owner"]).groupby("owner")["power_capacity_mw"]

plt.figure(figsize=(12, 7))
squarify.plot(sizes=owner_power.values, label=[f"{o}\n{p:.0f} MW" for o, p in ow
              alpha=0.8, pad=True)
plt.title("Provider Footprint by Total GPU Cluster Power Capacity (Top 12)")
plt.axis("off")
plt.tight_layout()
plt.show()
```



Finding: xAI's single build dwarfs every hyperscaler's entire footprint

xAI alone (783 MW, largely its Colossus site) exceeds Meta AI + CoreWeave + Google + Microsoft + Oracle combined (254 + 66 + 81 + 45 + 170 = 616 MW). A single concentrated build can outweigh several large, diversified providers at once.

Q10 — Can countries be grouped into AI infrastructure profiles based on energy, emissions, and capacity? (K-Means cluster plot)

Grouping the same 24 countries on 4 features: `renewables_share_elec`, `carbon_intensity_elec`, `log(total_power_mw)`, `log(clusters)` — standardized, then K-Means. Tried $k=2..6$ and picked by silhouette score.

```
In [23]: from sklearn.preprocessing import StandardScaler
from sklearn.cluster import KMeans
from sklearn.metrics import silhouette_score
```

```

features = country_sustainability[["renewables_share_elec", "carbon_intensity_el
features["total_power_mw"] = np.log1p(features["total_power_mw"])
features["clusters"] = np.log1p(features["clusters"])
X_scaled = StandardScaler().fit_transform(features)

for k in range(2, 7):
    labels = KMeans(n_clusters=k, n_init=10, random_state=0).fit_predict(X_scaled)
    print(f"k={k}    silhouette={silhouette_score(X_scaled, labels):.3f}")

kmeans = KMeans(n_clusters=3, n_init=10, random_state=0).fit(X_scaled)
country_sustainability["cluster"] = kmeans.labels_
country_sustainability.groupby("cluster").agg(
    n_countries=("clusters", "size"),
    avg_renewables_share_elec=("renewables_share_elec", "mean"),
    avg_carbon_intensity_elec=("carbon_intensity_elec", "mean"),
    total_power_mw=("total_power_mw", "sum"),
).round(1)

```

```

k=2    silhouette=0.385
k=3    silhouette=0.502
k=4    silhouette=0.415
k=5    silhouette=0.348
k=6    silhouette=0.328

```

```

Out[23]:

```

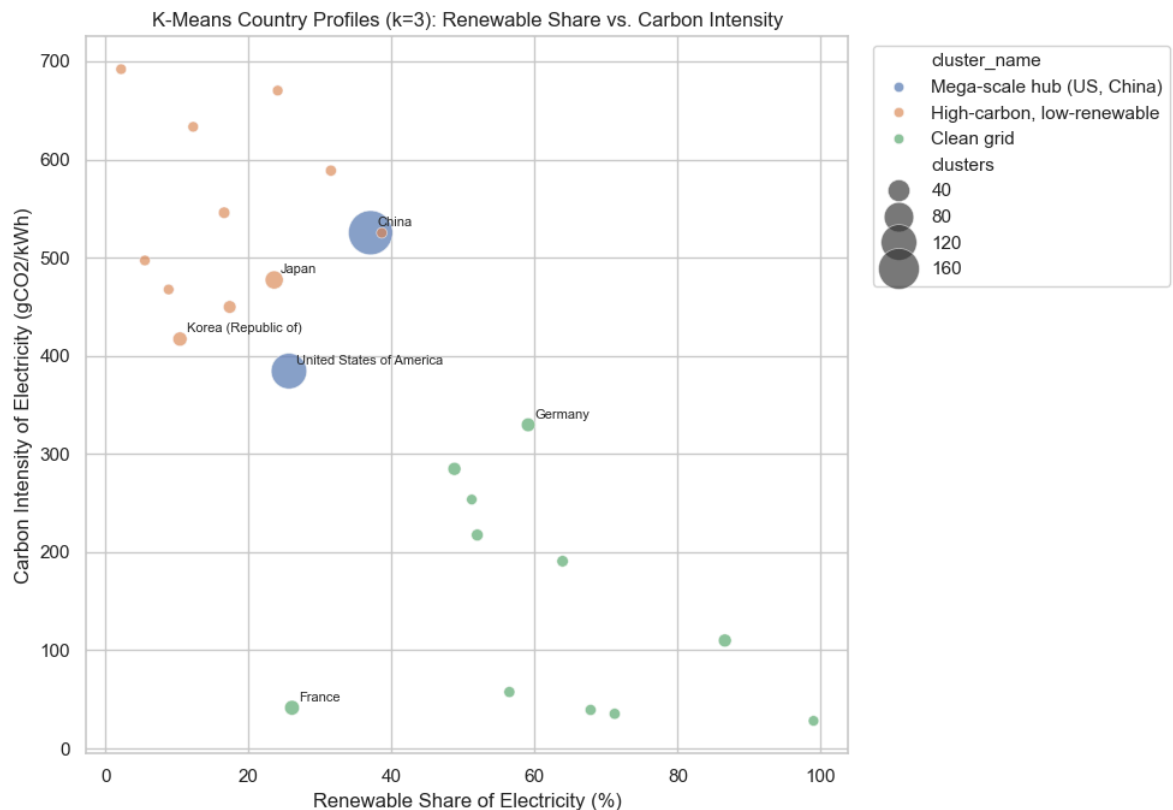
	n_countries	avg_renewables_share_elec	avg_carbon_intensity_elec	total_power_mw
cluster				
0	11	62.0	144.3	214
1	11	17.3	542.1	98
2	2	31.3	454.9	225

```

In [24]: # Name clusters from their own stats rather than hardcoding cluster IDs – KMeans
# numbering (0/1/2) is arbitrary and isn't guaranteed to match across runs.
cluster_stats = country_sustainability.groupby("cluster")[["renewables_share_elec", "carbon_intensity_elec", "total_power_mw", "clusters"]]
mega_scale_id = cluster_stats["total_power_mw"].idxmax()
remaining = cluster_stats.drop(mega_scale_id)
clean_id = remaining["renewables_share_elec"].idxmax()
dirty_id = remaining["renewables_share_elec"].idxmin()
cluster_names = {mega_scale_id: "Mega-scale hub (US, China)", clean_id: "Clean growth", dirty_id: "Polluting"}
country_sustainability["cluster_name"] = country_sustainability["cluster"].map(cluster_names)

plt.figure(figsize=(10, 7))
sns.scatterplot(
    data=country_sustainability, x="renewables_share_elec", y="carbon_intensity_elec",
    hue="cluster_name", size="clusters", sizes=(40, 700), alpha=0.65,
)
for country, row in country_sustainability.iterrows():
    if row["cluster"] == mega_scale_id or row["clusters"] >= 10:
        plt.annotate(country, (row["renewables_share_elec"], row["carbon_intensity_elec"]),
            fontsize=8, xytext=(4, 4), textcoords="offset points")
plt.title("K-Means Country Profiles (k=3): Renewable Share vs. Carbon Intensity")
plt.xlabel("Renewable Share of Electricity (%)")
plt.ylabel("Carbon Intensity of Electricity (gCO2/kWh)")
plt.legend(bbox_to_anchor=(1.02, 1), loc="upper left")
plt.tight_layout()
plt.show()

```

Finding: scale, not sustainability, is what actually separates countries

k=3 gave the best silhouette score (0.502) of the range tested, and the resulting groups are easy to name:

- **Mega-scale hub** — just US and China. They separate from everyone else almost entirely on raw scale (2,260 MW combined vs. ~100-215 MW for the other two groups), not on their sustainability numbers, which are mid-pack (26-37% renewable, 384-525 gCO₂/kWh).
- **Clean grid** (11 countries: Norway, Sweden, Switzerland, France, Finland, Brazil, Canada, UK, Netherlands, Italy, Germany) — avg 62% renewable, avg 144 gCO₂/kWh.
- **High-carbon, low-renewable** (11 countries: South Korea, Russia, UAE, Japan, Singapore, Australia, Thailand, Poland, Taiwan, India, Saudi Arabia) — avg 17% renewable, avg 542 gCO₂/kWh.

The takeaway for the "where should AI expand sustainably?" question: today's two largest AI build-out countries aren't chosen for grid cleanliness — they're just where the scale already is. If sustainability were the deciding factor, the "Clean grid" group (mid-size activity, 6-14 clusters each) is where new capacity would do the least carbon damage per MW.